

Supplementary information: rollback-consistent state transactions and auditable verification of an Abaqus multiphase poromechanics user element

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1. Evidence boundary and reproducibility objects

This supplement reports the complete registered P1 definitions and the negative evidence needed to interpret the main article. It does not add a geotechnical validation problem, mesh or time-step convergence, native balance closure, or a full-domain result. The public v0.8.2 artifact at <https://github.com/yqma7980/rollback-consistent-multiphase-uel-verification>, archived under 10.5281/zenodo.21773371, contains the frozen UEL and standalone harness, input state and direction manifests, preregistrations, complete CSV evidence used in all figures, plotting scripts, and SHA-256 manifests. Code uses the MIT License; data, documentation, figures, and manuscript-supporting material use Creative Commons Attribution 4.0 International.

The global sign convention is

$$\mathbf{RHS} = -\mathbf{R}, \quad D(\mathbf{RHS})[\mathbf{d}] \simeq -\mathbf{J}_{phys}\mathbf{d}. \quad (1)$$

The host receives $\mathbf{J}_{returned}$, while the registered physical-derivative test removes the unpaired matrix guard. No block, state, direction, or perturbation level selects an independent sign.

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2. Complete implementation definition

2.1. Parameters and frozen closure choices

Table S1: Complete registered PROPS map.

No.	Symbol or code role	Registered value	Unit	Evidence boundary
1	E	2.50×10^{10}	Pa	isotropic plane-strain elasticity
2	ν	0.300	–	isotropic plane-strain elasticity
3	α	1.00	–	equivalent-pressure coupling
4	$c_{p,\text{impl}}$ STORW	/ 1.00×10^{-6}	Pa^{-1}	implementation-level coefficient; legacy di- mensionless comment is metadata error; calibration/sensitivity pending
5	μ_w	1.00×10^{-3}	Pa s	wetting mobility
6	ρ_n	700	kg m^{-3}	non-wetting mass bal- ance
7	μ_n	1.00×10^{-3}	Pa s	non-wetting mobility
8	S_{wr}	0.200	–	residual wetting satu- ration
9	S_{nr}	0.0500	–	residual non-wetting saturation
10	p_b	1.00×10^4	Pa	entry-pressure parame- ter
11	λ_{BC}	2.00	–	Brooks–Corey ex- ponent (Brooks and Corey, 1964)
12	n_w	2.00	–	wetting relative- permeability exponent
13	n_n	2.00	–	non-wetting relative- permeability exponent

No.	Symbol or code role	Registered value	Unit	Evidence boundary
14	β_ϕ	1.00×10^{-2}	—	strain-saturation coupling coefficient
15	standalone debug seed	0.100	—	no physical role in registered production profile
16	K_0	6.00×10^{-12}	m ²	scalar absolute permeability
17	σ'_0	0	Pa	permeability reference stress
18	a_σ	0 or 1.00×10^{-8}	Pa ⁻¹	zero in non-stress-sensitive states; nonzero in registered stress-sensitive stages
19	CB_0	0	—	branch factor reference
20	a_{cb}	0	—	makes $s_{cb} = 1$ in registered profile
21	K_{min}	1.00×10^{-20}	m ²	lower hard bound
22	K_{max}	1.00×10^{-8}	m ²	upper hard bound
23	mechanics-state freeze switch	0	—	disabled diagnostic/control
24	state-writeback freeze switch	0	—	disabled diagnostic/control
25	mechanics-only compression switch	0	—	disabled diagnostic/control
26	strict mechanics compression switch	0	—	disabled diagnostic/control
27	no-writeback compression switch	0	—	disabled diagnostic/control

The four-node element is plane strain, uses 2×2 Gauss integration and

unit thickness in the P1 tests, and orders each node as (u_x, u_y, p_w, S_n) . The implemented closures are

$$S_w = 1 - S_n, \quad S_e = \text{clip}\left(\frac{S_w - S_{wr}}{1 - S_{wr} - S_{nr}}; \varepsilon, 1 - \varepsilon\right), \quad (2)$$

$$p_c = p_b(S_e^{-1/\lambda_{BC}} - 1), \quad k_{rw} = S_e^{n_w}, \quad k_{rn} = (1 - S_e)^{n_n}, \quad (3)$$

$$p_{eq} = p_w + (1 - S_e)p_c, \quad (4)$$

$$K_{raw} = K_0 e^{a_\sigma(\sigma'_m - \sigma'_0)} s_{cb}, \quad (5)$$

$$s_{cb} = 1 + a_{cb}(CB_{GP} - CB_0), \quad (6)$$

$$K = \text{clip}(K_{raw}; K_{min}, K_{max}). \quad (7)$$

They are frozen engineering choices, not a thermodynamic derivation or project calibration.

2.2. Weak residuals and nine-block derivative ledger

With current trial superscript k and committed beginning-state superscript n ,

$$\mathbf{R}_u = \int B^T \boldsymbol{\sigma}' d\Omega - \int B^T \alpha p_{eq} \mathbf{m} d\Omega, \quad (8)$$

$$R_{p,i} = \int \left[N_i c_{p,\text{impl}} \frac{p_w^k - p_w^n}{\Delta t} + N_i \alpha \frac{\varepsilon_v^k - \varepsilon_v^n}{\Delta t} + \nabla N_i \cdot (\lambda_w \nabla p_w^k) \right] d\Omega, \quad (9)$$

$$R_{s,i} = \int \left[N_i \phi \rho_n \frac{S_n^{use} - S_n^n}{\Delta t} + N_i \rho_n \beta_\phi S_n^{use} \frac{\varepsilon_v^k - \varepsilon_v^n}{\Delta t} + \nabla N_i \cdot \left\{ \lambda_n [\nabla p_w^k + p'_c(S_n^k) \nabla S_n^k] \right\} \right] d\Omega. \quad (10)$$

where $\lambda_w = K k_{rw} / \mu_w$ and $\lambda_n = \rho_n K k_{rn} / \mu_n$. RP is a wetting pressure-storage/volumetric balance and RS is a non-wetting mass balance; their native units differ and direct summation is forbidden.

Table S2: Equation-term-routine-block-unit-test traceability ledger.

Block	Rows/cols.	Differentiated terms	residual	Principal source routines	Native block unit	Verification evidence	evi-
K_{uu}	R_u/u	effective-stress force		UPS_MECH_CONSTITUTIVE; UPS_ASSEMBLE_RU_KU	force/length	mechanics-row FD; affine patch	
K_{up}	R_u/p_w	$-B^T \alpha p_{eq} m$ $\partial p_{eq}/\partial p_w$	through	UPS_TWOPHASE_CLOSURE; UPS_ASSEMBLE_RU_KU	force/pressure	mechanics-row FD	
K_{us}	R_u/S_n	equivalent-pressure saturation derivative		same as K_{up}	force/saturation	branchwise mechanics-row FD	
K_{pu}	R_p/u	strain coupling and stress-permeability wetting-flux chain		UPS_RESERVOIR_UPDATE; UPS_KABS_SIGMA_TO_U; UPS_ASSEMBLE_RP_KP	RP/ u	stress-sensitive FD	
K_{pp}	R_p/p_w	pressure storage and wetting diffusion		UPS_ASSEMBLE_RP_KP	RP/pressure	physical/numerical separation FD	
K_{ps}	R_p/S_n	wetting storage and mobility dependence		closure plus UPS_ASSEMBLE_RP_KP	RP/saturation	six-block regression FD	
K_{su}	R_s/u	strain coupling and stress-permeability non-wetting-flux chain		reservoir and UPS_ASSEMBLE_RS_KS routines	RS/displacement	stress-sensitive FD	
K_{sp}	R_s/p_w	non-wetting pressure-gradient flux		UPS_ASSEMBLE_RS_KS	RS/pressure	nullspace-aware and non-null FD	
K_{ss}	R_s/S_n	storage-use, strain, mobility, p'_c , and p''_c terms		closure; UPS_PC_SECOND_DERIVATIVE; UPS_ASSEMBLE_RS_KS	RS/saturation	smooth and stable-active-branch FD	

The only unpaired matrix term in the registered profile is $10^{-12}I_{16}$. It is mixed-dimensional in physical coordinates and becomes $W(10^{-12}I_{16})X$ after row/variable scaling. It is not a residual derivative, inf-sup stabilization, conservation device, or bound-preserving scheme.

2.3. Complete state-slot inventory and host ownership

The Abaqus UEL interface documents **SVAR**S as solver-managed state for general nonlinear increments and identifies **KINC=0** as a zero-increment output call (Dassault Systèmes SIMULIA, 2026). The registered source assigns 48 slots per Gauss point and 192 per element.

Table S3: All 48 Gauss-point state slots.

Slot	Quantity	Unit	Ownership	Role in registered source
1	p_w	Pa	physical/closure candidate	primary reconstruction; physical path
2	S_n	–	physical/closure candidate	primary reconstruction; physical path
3	S_w	–	physical/closure candidate	closure mirror
4	p_c	Pa	physical/closure candidate	capillary closure mirror
5	p_{eq}	Pa	physical/closure candidate	mechanics closure mirror
6	ε_v	–	physical/closure candidate	volumetric-strain history
7	ϕ	–	physical/closure candidate	porosity mirror
8	K	m ²	physical/closure candidate	absolute-permeability mirror
9	C_f	source-defined	physical/closure candidate	capacity value; provenance limited
10	σ'_{xx}	Pa	physical/closure candidate	effective-stress history
11	σ'_{yy}	Pa	physical/closure candidate	effective-stress history
12	σ'_{xy}	Pa	physical/closure candidate	effective-stress history
13	CAP_REF	Pa	committed/candidate physical history	authoritative capillary reference
14	CAP_REF_VALID	–	committed/candidate physical history	authoritative validity flag
15	DBG_DETJ	m ²	conditional diagnostic candidate	written only under DBG_MECH
16	DBG_EPSXX	–	conditional diagnostic candidate	written only under DBG_MECH
17	DBG_EPSYY	–	conditional diagnostic candidate	written only under DBG_MECH
18	DBG_GXY	–	conditional diagnostic candidate	written only under DBG_MECH
19	DBG_SIGXX	Pa	conditional diagnostic candidate	written only under DBG_MECH
20	DBG_SIGYY	Pa	conditional diagnostic candidate	written only under DBG_MECH

Slot	Quantity	Unit	Ownership	Role in registered source
∞	21 DBG_SIGXY	Pa	conditional diagnostic candidate	written only under DBG_MECH
	22 DBG_SM	Pa	conditional diagnostic candidate	mean-stress diagnostic
	23 DBG_RU5	row-dependent	conditional diagnostic candidate	mechanics residual diagnostic
	24 DBG_RU8	row-dependent	conditional diagnostic candidate	mechanics residual diagnostic
	25 DBG_K55	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	26 DBG_K88	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	27 DBG_K51	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	28 DBG_K55ROW	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	29 DBG_K58	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	30 DBG_K81	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	31 DBG_K85	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	32 DBG_K88ROW	entry-specific	conditional diagnostic candidate	matrix-entry diagnostic
	33 OUT_PW	Pa	conditional output mirror	current/core/prior-output source
	34 OUT_SN	–	conditional output mirror	current/core/prior-output source
	35 OUT_PC	Pa	conditional output mirror	current/core/prior-output source
	36 OUT_PEQ	Pa	conditional output mirror	current/core/prior-output source
	37 OUT_KABS	m ²	conditional output mirror	current/core/prior-output source
	38 OUT_EPSV	–	conditional output mirror	current/core/prior-output source

Slot	Quantity	Unit	Ownership	Role in registered source
39	OUT_SIGXX	Pa	conditional output mirror	current/core/prior-output source
40	OUT_SIGYY	Pa	conditional output mirror	current/core/prior-output source
41	OUT_SIGXY	Pa	conditional output mirror	current/core/prior-output source
42	OUT_VALID	–	output provenance	output-bundle validity
43	OUT_SOURCE	–	output provenance	source identifier
44	OUT_TIME	s	output provenance	total-time tag
45	OUT_KINC	–	output provenance	increment tag
46	reserved	–	reserved	initialized to zero; no other source reference
47	reserved	–	reserved	initialized to zero; no other source reference
48	reserved	–	reserved	initialized to zero; no other source reference

The source initializes slots 15–48 to zero on each candidate rebuild, then conditionally overwrites slots 15–32 under the mechanics-debug gate and slots 33–45 under the output-source gate. Static source tracing found no use of slots 15–45 in `RHS`, `AMATRIX`, or the construction of slots 1–14; the machine-readable audit records every read/write anchor. This establishes a static no-reverse-physical-path result, not universal dynamic noninterference. The repair stores `CAP_REF` and `CAP_REF_VALID` in slots 13–14. A candidate may be returned during a trial, but no private user-memory commit is authoritative. The tested contract is:

1. equal declared state after clean and permuted rejected histories;
2. clean versus forced-cutback accepted path;
3. zero-increment and terminal-call noninterference;
4. output-on/output-off physical-state parity;
5. independent-process replay and restart where registered.

Only these paths are covered; all-state and all-host-path safety are not claimed.

3. Registered geometry, states, directions, and numerical gates

3.1. Single-element and assembled-patch geometry

The single element has nodes $(0, 0)$, $(1, 0)$, $(1, 1)$, and $(0, 1)$. The assembled patch spans $[0, 2] \times [0, 2]$ with nine nodes and elements $(1, 2, 5, 4)$, $(2, 3, 6, 5)$, $(4, 5, 8, 7)$, and $(5, 6, 9, 8)$. All use four Gauss points, unit thickness, $\Delta t = 60$ s, `NDOFEL`=16, `NSVARS`=192, `PROPS`=27, `JPROPS`=1, and an unsymmetric returned matrix.

The affine single-element states are: *I0*: $u_x = 10^{-6}x$, $u_y = -5 \times 10^{-7}y$, $p_w = 1.20 \times 10^7 + 10^3x + 5 \times 10^2y$, and $S_n = 0.25 + 10^{-3}x + 5 \times 10^{-4}y$.

I1: $u_x = 10^{-5}x + 2 \times 10^{-6}y$, $u_y = -8 \times 10^{-6}y$, $p_w = 1.20 \times 10^7 + 2 \times 10^5x - 10^5y$, and $S_n = 0.30 + 5 \times 10^{-3}x$.

I2: $u_x = 5 \times 10^{-4}x + 10^{-4}y$, $u_y = -3 \times 10^{-4}y + 5 \times 10^{-5}x$, $p_w = 1.20 \times 10^7 + 2.5 \times 10^5x - 1.5 \times 10^5y$, and $S_n = 0.25 + 0.12x + 0.04y$. The complete committed 192-slot states are distributed in `inputs/wp10_i6_frozen_states.csv` rather than rounded in print.

3.2. Direction vectors and scales

The characteristic scales are $X_u = 10^{-5}$ m, $X_p = 10^7$ Pa, and $X_s = 10^{-1}$. Table S4 gives the exact scaled components inherited by every registered mixed-direction test; no second normalization, orientation, or sign selection was allowed.

Table S4: Scaled 16-component mixed directions.

Local DOF	Field	M1	M2	M3
1	u	0.2041241452319315	0.2041241452319315	0.08948580545678715
2	u	0.2041241452319315	-0.2041241452319315	-0.246215637382763
3	p_w	0.2886751345948129	0.2439781005892371	0.2886751345948129
4	S_n	0.4458272494941172	0.2886751345948129	0.2886751345948129
5	u	0.2041241452319315	0.2041241452319315	-0.06251452700161604
6	u	0.2041241452319315	-0.2041241452319315	0.2041594094186132
7	p_w	-0.2886751345948129	0.2724656467277031	0.2886751345948129
8	S_n	0.06155965610544123	0.2886751345948129	-0.2886751345948129
9	u	0.2041241452319315	0.2041241452319315	-0.01191099614095596
10	u	0.2041241452319315	-0.2041241452319315	-0.2614054454923604
11	p_w	0.2886751345948129	0.09584463572960104	0.2886751345948129
12	S_n	-0.1602936386585785	0.2886751345948129	0.2886751345948129
13	u	0.2041241452319315	0.2041241452319315	0.3874856694133703
14	u	0.2041241452319315	-0.2041241452319315	-0.02223804308012635
15	p_w	-0.2886751345948129	0.4363304904786234	0.2886751345948129
16	S_n	0.3241724156797066	0.2886751345948129	-0.2886751345948129

3.3. Perturbations, gates, and nonlinear benchmarks

Table S5: Frozen numerical gates and benchmark definitions.

Object		Registered definition
Central-difference levels		$10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}, 10^{-5}, 3 \times 10^{-6}, 10^{-6}, 3 \times 10^{-7}, 10^{-7}$; all retained
Integrated derivative	nonzero	relative $L_2 \leq 10^{-5}$ and cosine ≥ 0.99999
Patch derivative	nonzero	relative $L_2 \leq 10^{-8}$ and cosine ≥ 0.999999
Null response		scaled derivative, matrix-vector, and mismatch $L_2/L_\infty \leq 10^{-10}$
Parity/repeatability		exact when available; otherwise normalized maximum difference $\leq 10^{-13}$
Branch rule		fixed active branch; branch crossings retained but excluded from smooth claim; branch-distance factor > 10 where registered
N1		target I1; direction M2; primary offset 0.75; near offset 0.15
N2		target I2; direction M3; primary offset 1.00; near offset 0.20
Newton constraints		fixed local DOFs 1,2,3,4,6; free DOFs 5,7,8,9,10,11,12,13,14,15,16; maximum 20 updates
Newton order		at least five residual states; three consecutive $q \geq 1.8$; quadratic-constant ratio ≤ 10 ; final residual $\leq 10^{-12}$; root error $\leq 10^{-10}$; no globalization or branch crossing
Local-basin tudes	ampli-	21 values from 10^{-2} to 10^{-6} ; fixed bands L2 $[10^{-2}, 10^{-3}]$, L3 $[10^{-3}, 10^{-4}]$, L4 $[10^{-4}, 10^{-5}]$, L5 $[10^{-5}, 10^{-6}]$

Object	Registered definition
Local-basin order	root and residual OLS slopes in $[1.8, 2.2]$, fixed adjacent-slope and quadratic-constant gates, branch and resolution gates, and persistent entry in at least two adjacent registered bands

4. Complete supplementary evidence figures

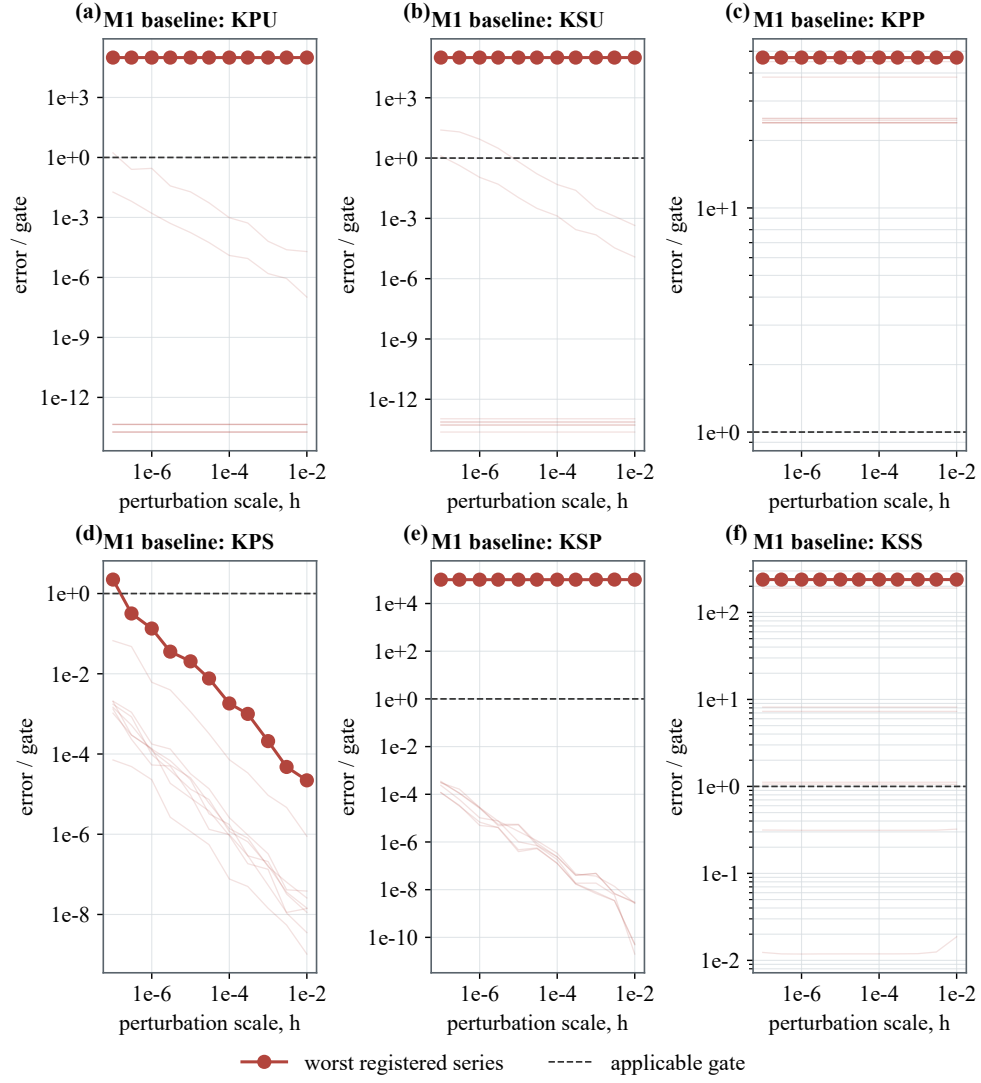


Figure S1: Complete M1 six-block baseline finite-difference curves. The baseline contains one consistent block, registered omissions, bounded approximations, a matrix-only pressure mismatch, and a uniform-pressure null response.

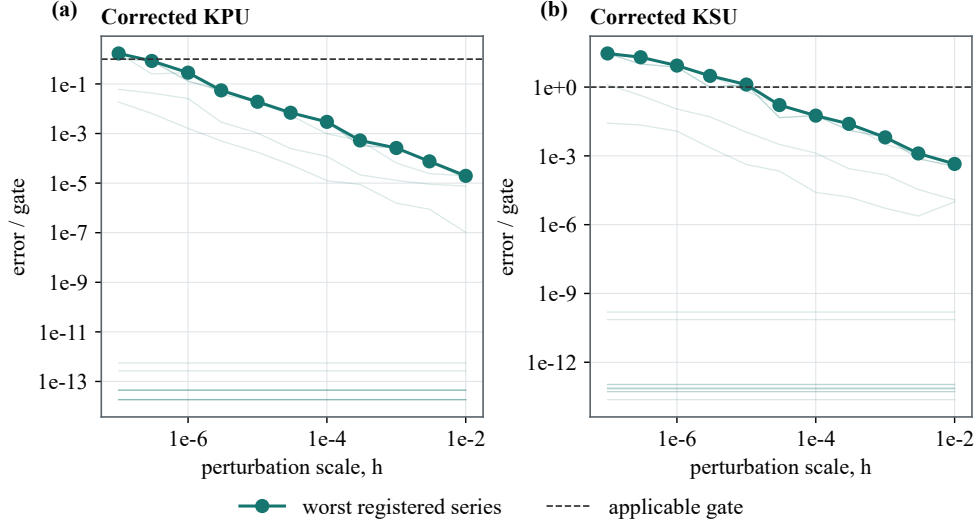


Figure S2: Stress-dependent permeability-chain corrections for K_{pu} and K_{su} across the retained perturbation levels.

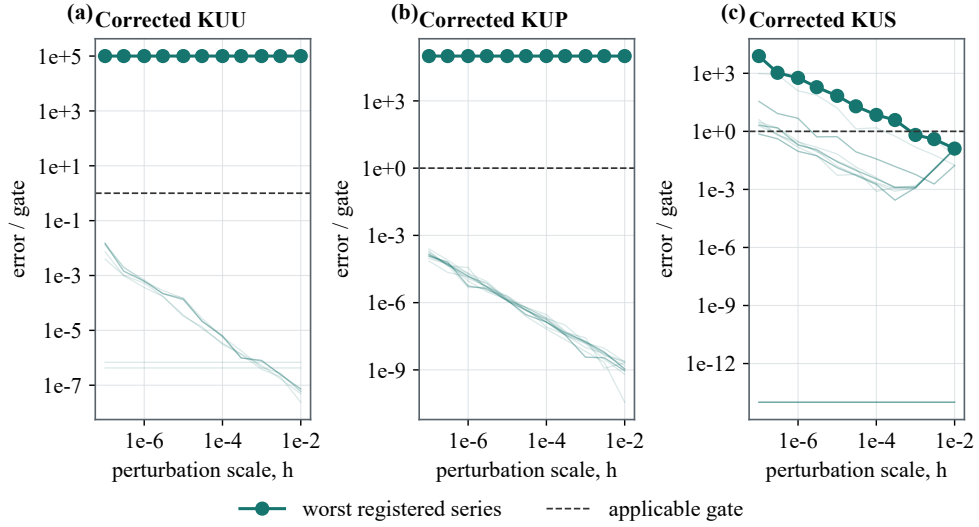


Figure S3: Mechanics-row derivative evidence for K_{uu} , K_{up} , and K_{us} .

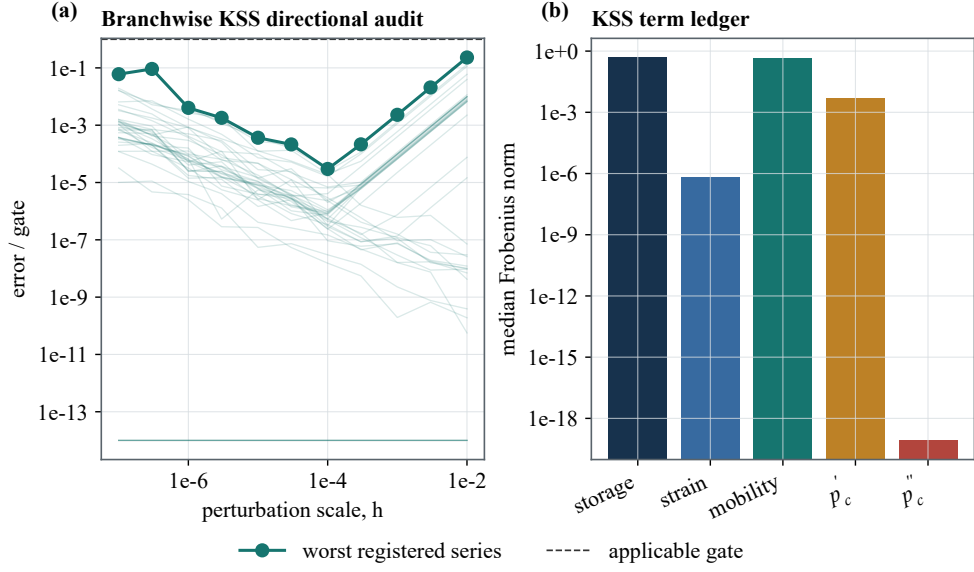


Figure S4: Smooth and stable-active-branch K_{ss} evidence and the storage, strain, mobility, first-capillary, and second-capillary derivative closure. Exact switching points are excluded.

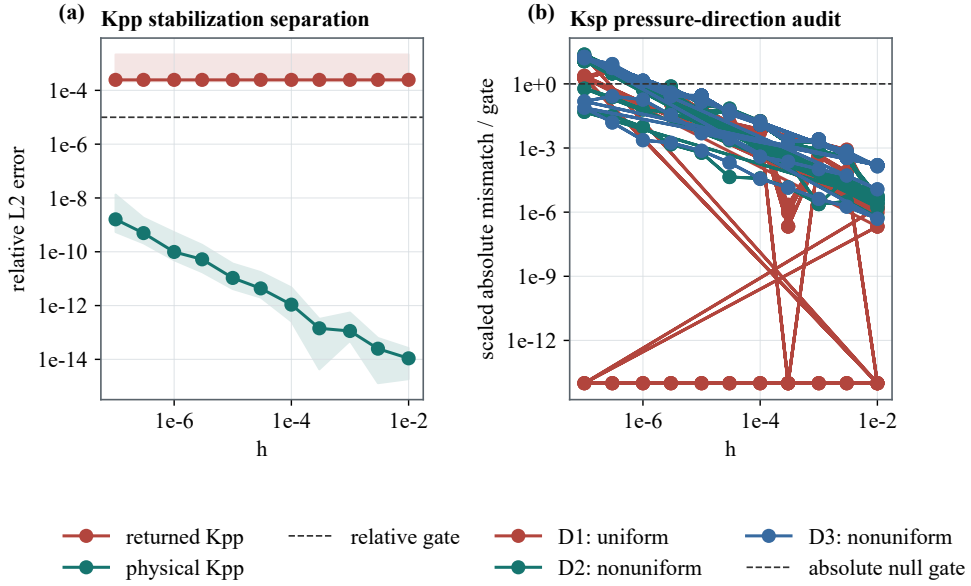


Figure S5: K_{pp} raw/physical/numerical separation and nullspace-aware K_{sp} evidence. The unscaled numerical diagonal is not a dimensional stabilization law.

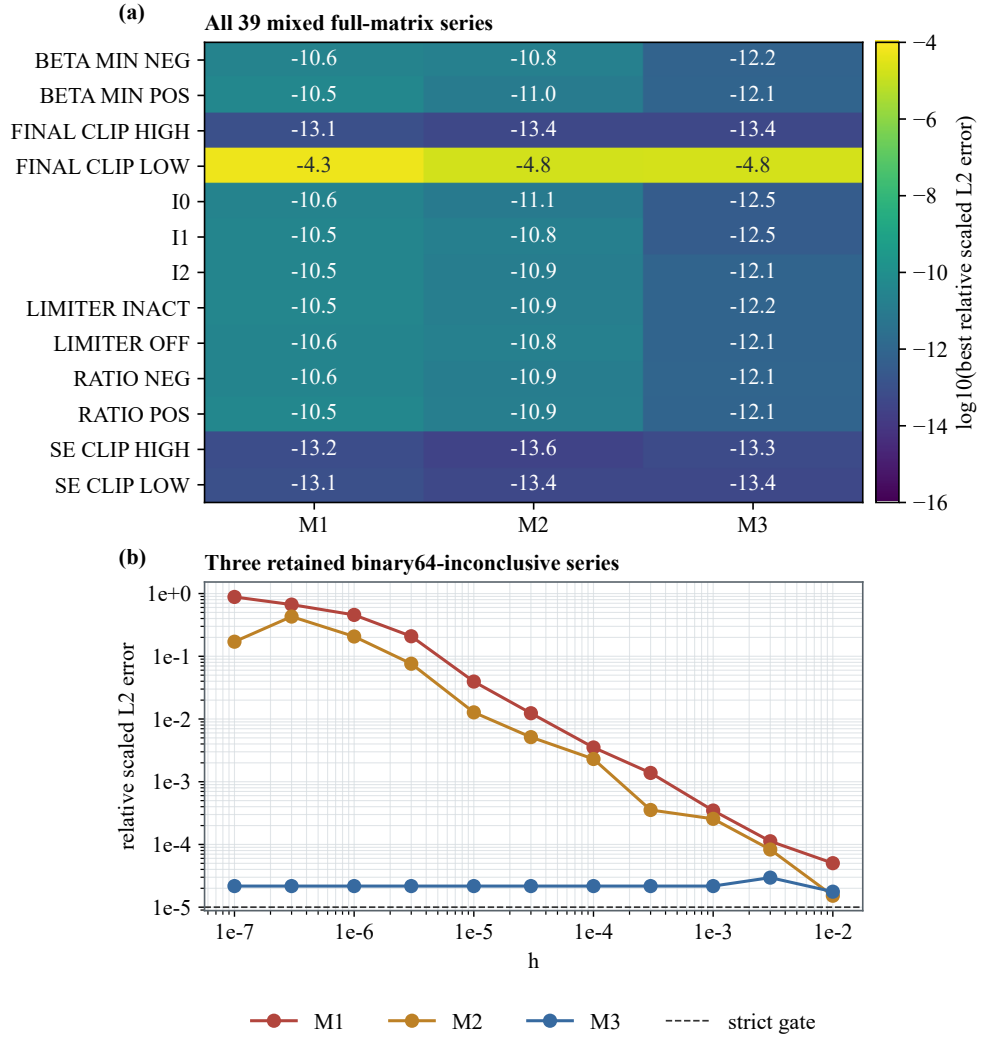


Figure S6: All 39 registered mixed full-matrix series. The strict binary64 record is 36/39 and remains unchanged.

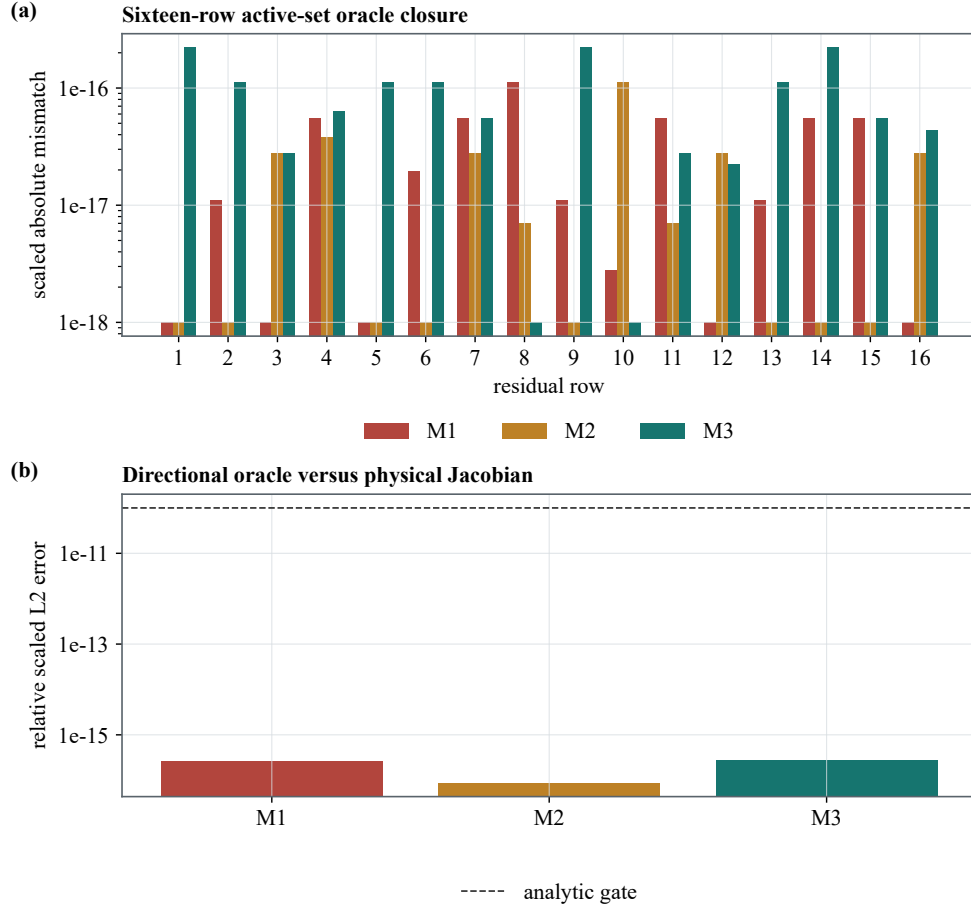


Figure S7: Sixteen-row independent stable-active-set derivative closure for the three lower-clip directions. This supplements but does not rewrite the strict binary64 record.

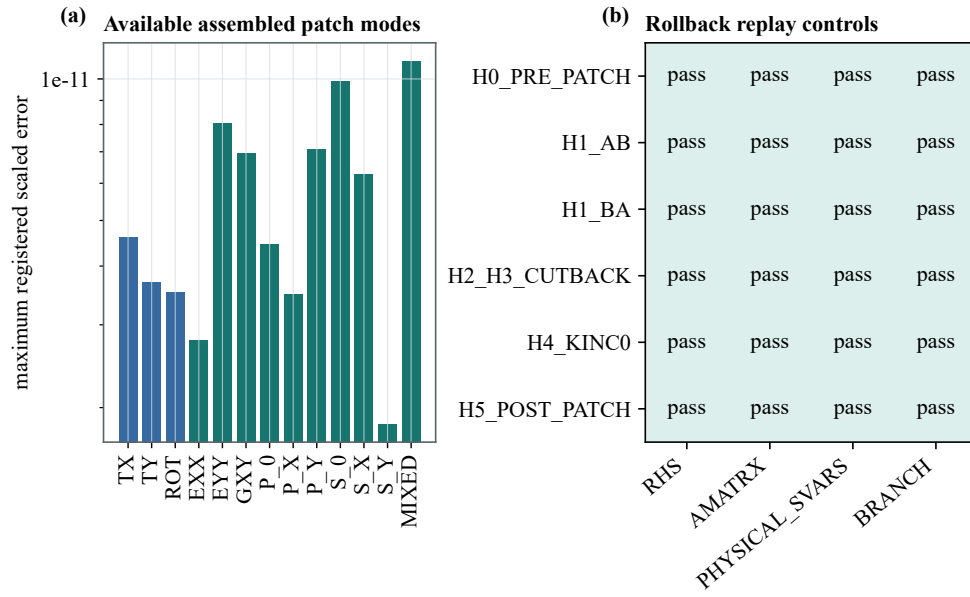


Figure S8: Available assembled directional-patch and rollback-replay controls. Term-by-term independent assembly closure and the second complete process were not executed after the Newton-order hard stop.

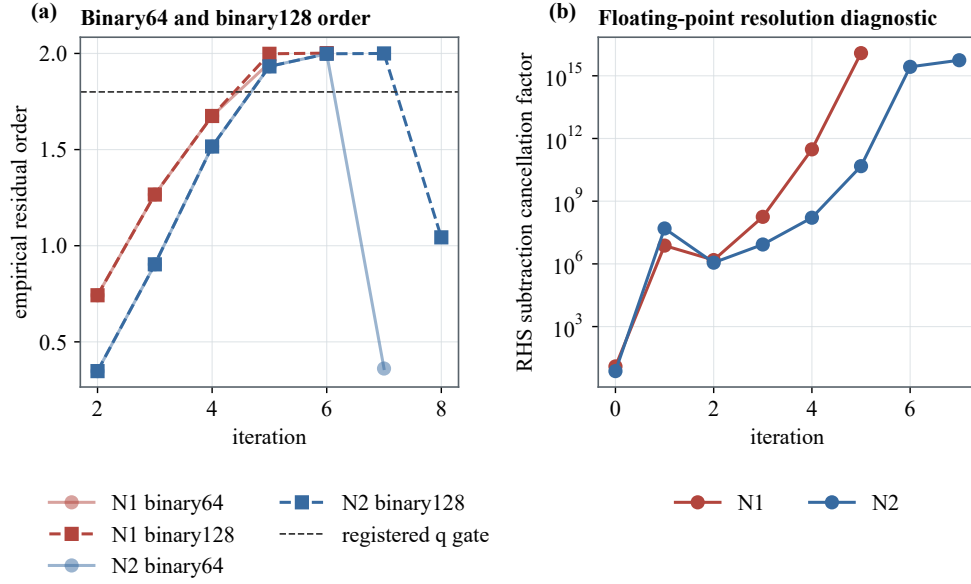


Figure S9: Binary64/binary128 residual-floor and empirical-order audit. Favorable N2 behavior does not override insufficient N1 evidence.

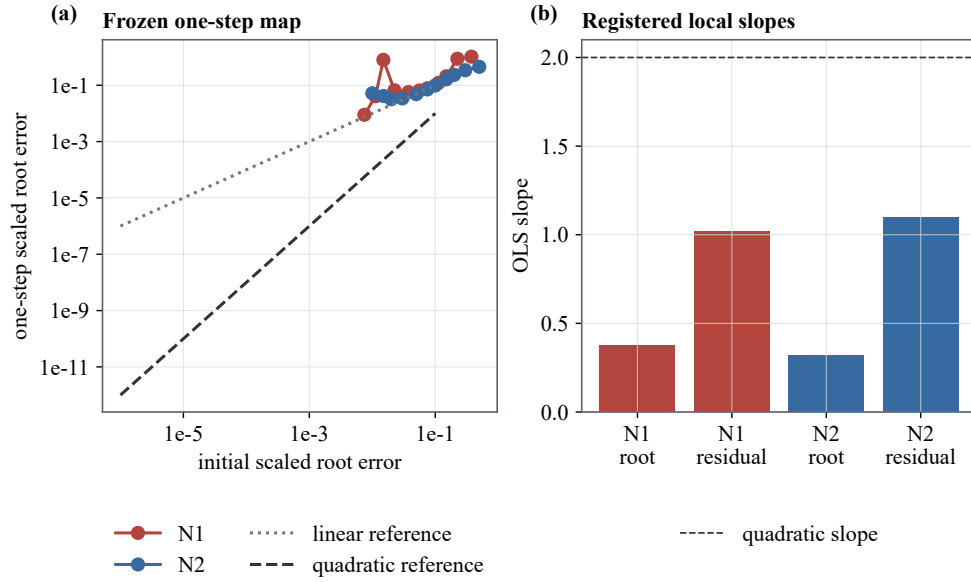


Figure S10: Frozen one-step-map root and residual slopes. The registered uniformly quadratic criterion was not met.

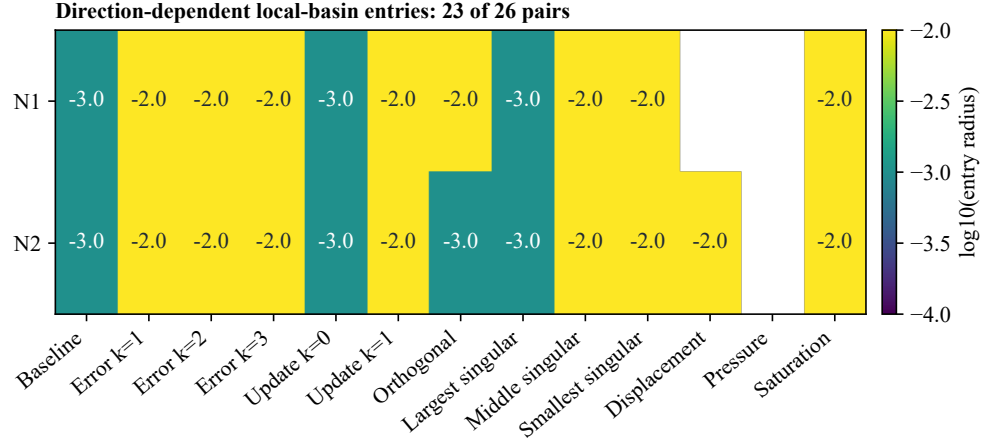


Figure S11: Corrected local-basin map. Persistent entry was observed for 23/26 benchmark-direction combinations; “no entry” is retained for the three field-only directions.

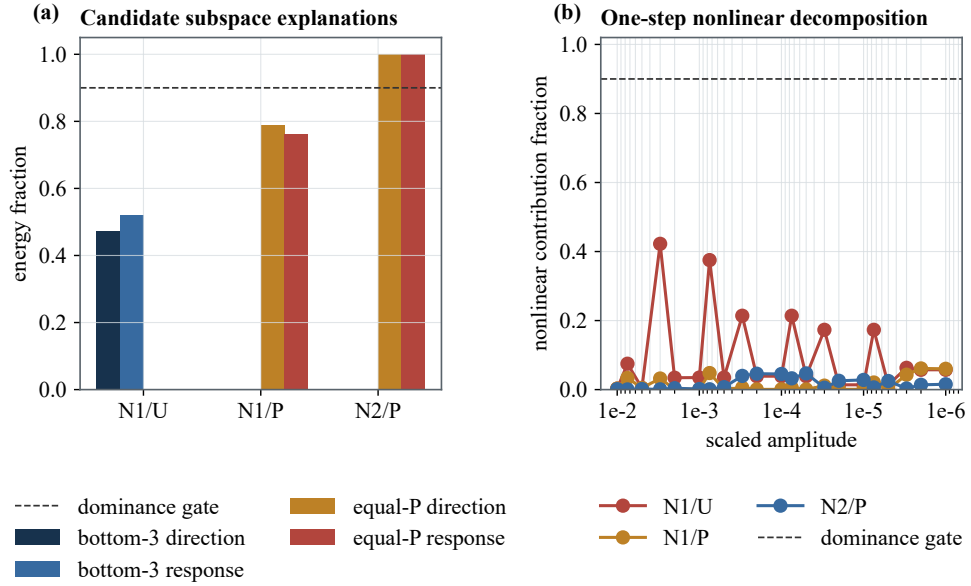


Figure S12: Projection, singular-subspace, and one-step-decomposition evidence for the three missing field directions. No single mechanism reached the frozen 90% explanation gate.

5. Negative and unresolved evidence retained

Table S6: Negative or incomplete verification outcomes that bound the manuscript claim.

Evidence layer	Retained outcome	Consequence
Strict mixed FD	36/39 binary64 series pass	active-set/binary128 supplement is not reported as 39/39 strict pass
Assembled patch	available run1 directional and replay gates pass	independent term assembly closure, M1/M2 parity, and full run2 were not completed after the hard stop
Newton order	N1 has zero and N2 has two rather than three consecutive eligible $q \geq 1.8$ segments	no two-benchmark quadratic-convergence claim
Resolution audit	N2 shows a high-precision local quadratic trend; N1 remains insufficient	no unique resolution-limited explanation
Frozen local map	registered root slopes 0.374 (N1) and 0.318 (N2); residual slopes 1.018 and 1.097	uniformly quadratic one-step map not observed; cause not localized to a term
Local basin	23/26 persistent entries	no common radius
Field subspaces	N1/ u , N1/ p , N2/ p lack a single 90% explanation	mixed or unresolved; no gauge/null proof
Equal-order Q4	no inf-sup, checker-board, mesh/time, maximum-principle, or bound study	field stability and boundedness unestablished
Native balance	RP and RS use different units; accepted-increment boundary flux is unavailable	no total two-phase mass or full-domain conservation claim

Evidence layer	Retained outcome	Consequence
Accepted-state output	results-file/context studies stopped on infrastructure/record-semantics gates	no full-domain accepted-state claim; not interpreted as physical failure

6. Reproducibility-package layout

The v0.8 review package is self-contained for figure reconstruction and organized as follows:

- `src/`: frozen verified UEL and standalone integrated finite-difference harness;
- `inputs/`: exact state, direction, branch, constraint, and benchmark manifests;
- `evidence/`: every CSV consumed by the figure builders, including the seven inputs omitted from v0.7;
- `preregistration/`: the principal frozen protocols;
- `scripts/`: package-relative main/supplementary figure builders and an isolated smoke test;
- `host_evidence/`: available repaired-build WP08 decks, launcher logs, parsed host outputs, and audits;
- `scripts/`: package-relative main and supplementary figure builders plus the clean-room smoke test;
- `MANIFEST_SHA256.csv`: relative path, byte count, and SHA-256 for every distributed object.

The figure-rebuild smoke test is executed from a fresh directory with no reference to the author workspace and checks all 54 PDF/SVG/PNG figure outputs against expected hashes. The package does not include Abaqus executables or licensed solver binaries, and the available WP08 host archive lacks the unsafe source and original `.sta/.msg/.dat` triplet needed for a complete before/after host replay. The public identifiers and licenses improve inspectability but do not upgrade the bounded lifecycle, derivative, convergence, conservation, or engineering-qualification claims.

References

- Brooks, R.H., Corey, A.T., 1964. Hydraulic Properties of Porous Media. Technical Report 3. Colorado State University. Fort Collins, Colorado. URL: <https://hdl.handle.net/10217/61288>.
- Dassault Systèmes SIMULIA, 2026. Abaqus 2026 User Subroutines Reference Guide: UEL. Dassault Systèmes. URL: <https://help.3ds.com/>. local installed documentation, SIMACAESUBRefMap/simasub-c-uel.htm.